PREPARED EXCLUSIVELY FOR CLIENT

مركز الربيع للأسنان
AL RABEEH DENTAL CENTER
Hamad Town W.L.L | Muharraq W.L.L | Isa Town W.L.L | Budaiya W.L.L | Galali W.L.L | Sitra W.L.L

ENTERPRISE HEALTHCARE DIGITAL TRANSFORMATION
Dark Blue & Orange Unified Clinical Theme (CORE-04)

ENGINEERED & PROPOSED BY

ساندس لاب الشرق الأوسط ذ.م.م
SaNDS LAB MIDDLE EAST W.L.L
@trichur infopark | since 2008 | an ISO 9001:2015 Certified Company

TECHNICAL ARCHITECTURE, BLUEPRINT AUDIT, MILESTONES & COMMERCIAL PROPOSAL

Al Rabeeh Dental Center — Custom EMRS & Reception Ecosystem Development

Comprehensive Engineering Analysis of the 35-Page *Reception Login Software Development Blueprint (Revised 19 Oct 2026)* — Featuring Advantages, Drawbacks, Strategic Enhancements, ReactJS + PHP MVC Microservices + MySQL + RabbitMQ Queuing + Custom .NET Smart CPR Driver + Offline SQLite/IndexedDB Auto-Sync, 6-Month Accelerated Delivery Schedule (+ 1-Month Deployment Buffer), Recalculated Commercials, and Governance Terms.

6 Bahrain Branches Supported | 6 Months Core + 1 Month Buffer Extra | Custom .NET Smart CPR Driver & Offline Sync | 100% Client Source Code Ownership

EXECUTIVE PROPOSAL CONTROL SUMMARY

Accelerated 6-Month Delivery + 1-Month Buffer

CLIENT ORGANIZATION	ENGINEERING PARTNER	SPECIFICATION BASELINE	AUTHENTICATED ROLES
Al Rabeeh Dental Center 6 Branches Across Kingdom of Bahrain	**SaNDS Lab Middle East W.L.L** ISO 9001:2015 Certified Division	**Part 1 Blueprint (35 Pages)** 11 Modules • 87 Explicit Rules	**Reception, Doctor & Admin** Includes Manager Governance Role
CORE TECHNOLOGY STACK	**6-MONTH CORE VALUE**	**TOTAL CONTRACT OFFER**	**IP & CONTRACT GOVERNANCE**
ReactJS + PHP MVC + MySQL RabbitMQ • .NET CPR Driver • SQLite Sync	**BD 7,200.000** 24 Working Weeks (6.00 Months)	**BD 8,400.000** 28 Working Weeks (7.00 Mos with Buffer)	**100% Client Source Code** Max 3 Revisions • 60-Day Notice

01 Executive Summary & Blueprint Architectural Audit

87 Explicit Functional Rules Audited (CORE, DASH, CAL, MSD, FV, ORTHO, TP, FB, LAB, INS, PJ, BILL)

SaNDS Lab Middle East W.L.L. has conducted an exhaustive line-by-line engineering analysis of **Al Rabeeh Dental Center's** *Reception Login Software Development Blueprint (Part 1, Revised 19 October 2026)*. The blueprint defines a mission-critical front-office, patient recall, dental lab coordination, insurance claim submission, clinical quality checklist, and cashier billing platform designed to unify operations across **Reception, Doctors, and Administration (including Manager functions)**.

FUNCTIONAL SCOPE

11 Modules

+ 9 Shared Core Rules (CORE-01 to 09)

PATIENT RECALL ENGINES

5 Trackers

MSD, Fail Visit, Ortho, TP & Feedback

RECEPTION KPI CARDS

6 Workstreams

Calls, Collection, IG, Google, Ortho/Dental

BAHRAIN LOCALIZATION

.000 BHD & CPR

Smart Card Reader + Globemed/Mednet

MANDATORY UNIFIED NAVIGATION SEQUENCE ACROSS ALL 3 LOGINS (CORE-02 PRECEDENCE RULE)

1. Dashboard (Default)
2. Calendar
3. MSD
4. Fail Visit
5. Ortho Tracker
6. TP Tracker
7. Feedback
8. Lab
9. Insurance
10. Patient Journey
11. Billing/Accounts

02 Blueprint Analysis: Advantages, Drawbacks & Strategic Enhancements

Deep Architectural & Operational Assessment

Our domain specialists evaluated the blueprint against modern GCC multi-branch dental clinic standards, Bahrain National Health Regulatory Authority (NHRA) norms, and high-concurrency web engineering practices. Below is our structured breakdown of the blueprint's inherent strengths, architectural drawbacks/gaps, and value-engineered enhancements.

1. Key Blueprint Advantages

Zero Patient-Leakage Architecture

Unlike passive clinic schedulers, the blueprint creates **four specialized safety nets** (*MSD, Fail Visit, Ortho Tracker, and TP Tracker*) that automatically surface every patient who left without a booking, missed yesterday's slot, is overdue for braces review, or has an unstarted treatment plan.

Verified Receptionist KPI Gamification

The 6-card Dashboard (Calls, Collections, Instagram, Google Reviews, Dental-to-Ortho, Ortho-to-Dental) separates **Admin Approval status** (Approved/Hold/Rejected) from **Target Achievement** (DASH-03), ensuring only audited conversions count toward staff incentives.

First-Class Bahrain Localization

Native handling of **Bahraini CPR smart-card readers** (preserving leading zeros as text per CORE-06), **3-decimal BHD arithmetic (.000)**, local payment rails (**Credimax, EazyPay, Cheque, Cash**), and Bahraini TPA claim formats (**Globemed & Mednet**).

Strict Domain Status Separation (CORE-07)

Explicitly separates appointment status, clinical procedure status, call outcome, task status, admin approval, and target achievement—preventing the state-corruption bugs common in poorly modeled clinic systems.

End-to-End 24-Point Quality Audit (PJ-03)

The **Patient Journey** module acts as a real-time clinical governance board for every Seen patient, visually flagging missing consents, before/after photos, screenings, or financial agreements in red.

Self-Aware Specification Hygiene

The blueprint proactively identifies Excel source flaws (e.g., `=TODAY()` misplaced next to File No., Ortho Tracker mislabeled as "Fail Visit", duplicate Calendar images) so engineers do not blindly replicate mockup errors.

2. Identified Drawbacks & Risks

Cross-Role Dependency Bottleneck (Part 1 Only)

The blueprint specifies Reception (Part 1) while deferring Doctor and Admin specs. However, **9 of 11 Reception modules cannot function without Doctor/Admin triggers** (e.g., Doctors marking procedures *In Progress/Completed* for MSD/TP/Feedback/Lab, Favorite TP flagging, and Admin approving the 6 KPI trackers).

Manual Double-Entry in Collection Tracker

In `DASH-02`, Reception manually enters Collection records (Doctor, Treatment, Paid, Balance) while also recording receipts in `BILL-01`. Without automated pre-fill from Billing, receptionists face double data-entry and reconciliation errors.

Undefined External Dental Lab Access (LAB-09)

The Lab module features a 3-lane chat (*Doctor, Reception, Lab*) and stage transitions (*Ready to Dispatch → Check Work → Work Ready*), yet `CORE-01` restricts logins strictly to Reception, Doctors, and Admin—leaving no authenticated channel for external lab technicians to respond.

Missing UI Specs & Status Master Conflicts

`FV-08` requires "Call List" to open a new window whose source Excel tabs (Sheet2/Sheet3) are completely blank. Furthermore, `ORTHO-07` has a conflicting status master in cells Q45:Q54 versus the UI buttons (e.g., *Debonded From Outside* vs. *Patient Did From Outside*).

Bulk WhatsApp Number-Ban Risk (CAL-11)

Sending bulk appointment reminders via unofficial WhatsApp web automation risks immediate Meta number banning for the clinic. Official Meta WhatsApp Cloud API template governance and rate-limiting are not specified.

Multi-Branch & Offline Continuity Gaps

Al Rabeeh operates **6 branches in Bahrain**, and front-desk reception cannot afford downtime during internet drops. The blueprint requires an offline-capable local store with automatic live-server sync.

3. Proposed Enhancements

Functional Doctor & Admin Bridge Console

Rather than building dead-end mocks, SaNDS Lab includes a **Shared Doctor & Admin Operational Console (Milestone 12)** so Doctors can update procedure statuses, tooth charts, lab orders, and comments, and Admins can approve KPI trackers and manage schedules from Day 1.

One-Click Billing-to-Collection Auto-Sync

When Reception generates a receipt in **Billing/Accounts (BILL-02)**, our engine automatically drafts the linked **Collection KPI entry (DASH-02)** using a shared immutable Receipt ID (CORE-09)—eliminating manual re-typing and preventing duplicate financial counting.

Interactive FDI Odontogram (Tooth Chart)

To power tooth-level recall rules in `TP-06` (Extraction w/o Implant), `TP-07` (RCT w/o Crown on same tooth), `LAB-04` (3-zone Shade Master), and `INS-03`, we embed a visual FDI Dental Tooth Selector (`#11–#48` & `#51–#85`).

Magic-Link External Lab Collaborator View

Solves `LAB-09` without adding a 4th top-level clinic login: external dental labs receive a secure, job-scoped magic link / PIN portal to post messages into the "Lab" chat lane and flag cases ready for clinic inspection.

Digital Tablet E-Consent & Photo Dropzone

Enhances **Patient Journey (PJ-03)** with a bilingual (Arabic/English) tablet signature pad for *General Consent, Treatment Consent & Financial Agreement*, turning red missing items green automatically upon patient signature.

Offline-First SQLite / IndexedDB Auto-Sync

Enables uninterrupted front-desk scheduling, patient lookup, and billing even during internet outages by caching encrypted records in local **SQLite / IndexedDB** and automatically synchronizing with the live PHP server upon reconnection.

03 Technology Stack, RabbitMQ Queuing, .NET Card Driver & Offline Sync

ReactJS + PHP MVC + MySQL + RabbitMQ + .NET CPR Service + SQLite/IndexedDB

All modules and shared interfaces of the Al Rabeeh Dental Center EMRS will be engineered strictly in accordance with the following enterprise technology stack, message-queuing middleware, custom .NET hardware driver, security encryption standards, and offline-resilient synchronization architecture:

Front-End (ReactJS) & Custom .NET Card Reader Service

Presentation & Hardware Layer

High-performance Single Page Application built on **ReactJS** paired with a custom **.NET Smart Card Driver & Windows Background Service**.

- **Unified Clinical Theme (`CORE-04`):** Strict Dark Blue & Orange design system across Reception, Doctor, and Admin views.
- **Custom .NET CPR Card Reader Driver (`CAL-09`):** Dedicated **.NET Driver & Windows Background Service** engineered to read Bahrain Smart CPR Cards directly from USB hardware and stream patient demographics into the ReactJS booking and patient file screens.
- **Real-Time UI State:** Instant counter refreshes across all 20 Calendar badges, 6 Dashboard KPI cards, and 24 Patient Journey columns.

PHP MVC Microservices, MySQL DB & RabbitMQ Queuing

Middleware, Database & Queue Layer

Modular **PHP MVC Microservices** backed by a **MySQL Relational Database** and **RabbitMQ Server Queuing Services**.

- **MySQL Back-End Database:** Enterprise **MySQL** database engine enforcing ACID transactions, strict `DECIMAL(12,3)` BHD precision, text-preserved CPR/File IDs (`CORE-06`), and immutable audit history (`CORE-08`).
- **RabbitMQ Server Queuing Services:** High-throughput **RabbitMQ** message broker handling asynchronous server queues for offline-to-live data synchronization, Bulk WhatsApp/SMS dispatch (`CAL-11`), midnight Fail Visit rollovers (`FV-01`), and idempotent transaction processing (`CORE-09`).
- **PHP MVC Microservices:** Domain-isolated services for Calendar, 5 Recall Trackers, Dental Lab Chat, TPA Insurance, and Cashier Billing.

Full Encryption & SQLite / IndexedDB Offline Sync

Resilience & Security Layer

End-to-end cryptographic security paired with an **Offline-Enabled Local Storage & Auto-Sync Engine**.

- **Enterprise Encryption (`CORE-08`):** TLS 1.3 (HTTPS) in transit, **AES-256 encryption at rest** for patient records, attachments & local cache, and Argon2/Bcrypt credential hashing.
- **SQLite / IndexedDB Offline Store:** During internet drops, all reception operations are stored locally in encrypted **SQLite / IndexedDB** tables.
- **Automatic Live-Server Sync via RabbitMQ:** Background sync workers detect internet restoration and automatically push queued offline records through **RabbitMQ** into the live **MySQL** server with conflict resolution.

END-TO-END HARDWARE, OFFLINE & RABBITMQ SYNCHRONIZATION PIPELINE

Combines local .NET CPR card reading and SQLite/IndexedDB offline storage with RabbitMQ server queuing and MySQL persistence.

1. .NET CPR Driver & ReactJS UI
2. Local SQLite / IndexedDB
3. RabbitMQ Queue Service
4. PHP MVC & Live MySQL DB

04 Resolution Matrix for Open Blueprint Conflicts & Ambiguities

Pre-Engineered Solutions for All Flagged Blueprint Items

The blueprint explicitly flags several conflicts between the source Excel sheets and reference mockups for confirmation before coding. To accelerate project kickoff, SaNDS Lab proposes the following best-practice engineering resolutions:

Rule Ref	Blueprint Conflict / Open Item	SaNDS Lab Recommended Engineering Resolution
DASH-04	**Multi-Day Filter & Default Till-Date Period** Source shows Today & Till Date equal on custom dates.	Default **Till Date** to *1st of Current Month → Today (MTD)*. When a user selects "Yesterday", Today shows Yesterday and Till Date shows Month-to-Yesterday. When a custom range is applied, both columns synchronize to the filtered range with dynamic header badges.
DASH-09	**Monthly Task Due Date on Short Months** E.g., task set for 31st in Feb/Apr/Jun/Sep/Nov.	Implement an automatic **Last-Day-of-Month Clamp Rule**: any recurring monthly task scheduled for day 29, 30, or 31 automatically triggers on the final calendar day of shorter months (e.g., Feb 28/29 or Apr 30).
DASH-12	**24-Hour Edit Window & Hold Record Clock** Clock start and handling of Held records after 24h.	Start the 24-hour countdown from `created_at` timestamp. If Admin places a record on **Hold** (orange), the 24-hour lock automatically re-opens for **an additional 24 hours from the Hold timestamp** so Reception can correct the record as required by `DASH-13`.
FV-03 / FV-08	**All-Doctors Date Filter Conflict & Blank Sheet2/3 Call List**	Show both *Filter-Scoped Count* and *Cumulative Unresolved Till-Date Count* in the All-Doctors header. Design the missing **Call List Window (FV-08)** as a slide-over drawer displaying full chronological call attempts, receptionist attribution, and text notes.
ORTHO-07	**Status Master Mismatch (Q45:Q54 vs. Buttons)** *Debonded From Outside* vs. generic Fail Visit list.	Implement a **Configurable Status Master per Module**: Ortho Tracker retains its clinical button statuses (including *Debonded From Outside*) as primary quick-filters while supporting secondary call sub-statuses (*Remind Later, SMS/WhatsApp*) in the dropdown.
TP-05..08	**Recall Interval Inclusivity & Tooth Matching** Exact 6m/2m vs. overdue; RCT w/o Crown tooth link.	Use **Overdue Inclusive Logic (`>= 6 Months` / `>= 2 Months`)** without subsequent recall completion so patients are not missed if reception checks a day late. Match *RCT Without Crown* and *Extraction Without Implant* by (`patient_id`, `fdi_tooth_number`).
FB-03 / PJ-05	**Feedback "Today vs. This Month" & PJ Complaint Columns**	Default Feedback to **Today** (per Excel rule) with a one-click toggle to "This Month". In Patient Journey, support configurable column visibility for *Initial Complaint, Chief Complaint, PFE (Patient Financial Estimate), and FLP (Follow-Up Plan Note)*.

05 Comprehensive Functional Breakdown (All 11 Blueprint Modules + Core)

100% Alignment with Pages 3–35 of Specification

1. Reception Dashboard & 6 KPI Trackers

DASH-01 to DASH-14

- **6 Independent KPI Cards:** Calls (with *Seen Pt* counter), Collections (BHD), Instagram, Google Review, Dental-to-Ortho, and Ortho-to-Dental conversions.
- **Dual Status Governance:** Admin Approval (Approved / Hold / Rejected) strictly decoupled from Target Achievement (Achieved / Not Achieved).
- **Visual Analytics:** 6-Month Target vs. Actual bar charts + gauge views with independent per-card date filters (From/To, Monthly, Weekly, Yesterday).
- **Task & Reminder Hub:** Admin-assigned Daily/Weekly/Monthly tasks (status-only for Reception) + Personal Task list + Hold/Rejection/Appointment-Due alerts + 24h audit-logged editing.

2. Multi-Doctor Calendar, CPR & Bulk WhatsApp

CAL-01 to CAL-12

- **Interactive Doctor Columns:** Doctor photo, duty hours, drag-and-drop slot rescheduling with confirmation modal, and drag-to-resize duration with out-of-hours warnings.
- **20 Live Status & Category Counters:** 16 appointment statuses (Scheduled, Pre-Confirmed, Confirmed, Coming, Arrived, In Room, Seen, No Ans 1/2, Fail Visit, etc.) + 4 patient counters.
- **Bahrain Smart CPR Card Reader:** One-tap hardware CPR extraction in both booking modal and patient file, with mandatory Bahrain phone format validation (CAL-10).
- **Omnichannel Messaging:** Bulk & individual WhatsApp, SMS, and Email dispatch with live delivery badges (Sent, Not Sent, Failed/Not Delivered) and Fail Visit rollover button.

3. Missing Schedule (MSD) Leakage Tracker

MSD-01 to MSD-05

- **Automated Unbooked Patient Detection:** Real-time query surfacing patients requiring further treatment, review, or balance payment who have no active future appointment.
- **7 Specialized Cohort Cards:** *In Progress Dental, In Progress Ortho, Decided Plan Dental, Decided Plan Ortho, Balance Payment Dental, Balance Payment Ortho, Completed With TP Ortho*.
- **Closed-Loop Conversion:** Displays Last Appt Date, File No., Photo, Responsible Receptionist, and Procedures; booking via the inline `Appointment` button auto-removes the patient and updates counts.

4. Fail Visit & No-Show Recovery Engine

FV-01 to FV-09

- **Idempotent Midnight / Button Rollover:** Yesterday's unattended appointments (Scheduled, Pre-Confirmed, Coming, Confirmed, Not Confirmed, No Ans 1/2) automatically transition to *Fail Visit* without duplicates.
- **Doctor & All-Doctors KPI Panels:** Tracks Total PT, Calls Done, and Pending (`Total PT - Calls Done`) with 10 rolling date presets (Today, Yesterday, Last 7/30 Days, Custom Range).
- **10 Outcome Filters & Smart Booking Lock:** Status buttons for *Schedule, No Ans 1/2/3, Remind Later, Call Back Us, Cancelled, SMS/WhatsApp, Outside, Not Aware TP*; disables booking button if future slot exists.

5. Orthodontic (Ortho) Continuity Tracker

ORTHO-01 to ORTHO-08

- **Running Braces Cohort Management:** Lists all patients with *In Progress* orthodontic procedures filtered by orthodontist with doctor photo and active patient counts.
- **5 Appliance Categories & Review Counters:** Expandable *Total Running PT* breakdown (Special, Normal, Invisalign, Illusion, Outside) alongside Today Review, Till Date Review, and Review Pending.
- **7 Overdue Aging Buckets:** Non-overlapping *1 Month, 2 Month, 3 Month, 4 Month, 5 Month, 6 Month, and Above 6 Months* cohorts based on last visit date with full call-list logging.

6. Treatment Plan (TP) & Recall Tracker

TP-01 to TP-10

- **Strict 3-Condition Inclusion Engine:** Surfaces patients who have (1) a pending Treatment Plan, (2) no treatment currently *In Progress*, and (3) no active future appointment booked.
- **Financial & Priority Tracking:** Displays *Total TP Value (BHD)* vs. *Total TP Value Done (BHD)* and consumes Doctor's **Favorite TP Patients** selection.
- **7 Clinical Recall & Upsell Cohorts:** *Periodic Scaling (6m), Periodic GBT (2m), Periodic Whitening (6m), Extraction Without Implant, RCT Without Crown (tooth-matched), Ortho, and Implant*.

7. Post-Treatment Feedback Call Tracker

FB-01 to FB-06

- **Automated Clinical Trigger:** Populates automatically when a doctor marks qualifying procedures *In Progress* or *Completed*.
- **Live Conversion Funnel:** Displays Total PT to Call, Called (Done %), No Ans 1 %, No Ans 2 %, No Ans 3 %, and Pending Calls %.
- **Dual-Role Comment Segregation:** Preserves view-only *Doctor Comments* alongside Reception follow-up notes (enabled upon selecting *Done*) and direct review appointment booking.

8. Dental Lab Workflow & 3-Lane Chat

LAB-01 to LAB-10

- **Detailed Prosthetic Job Order:** Captures Dental Lab name, Emergency/Trial/Call Doctor/Technician Appt flags, Upper/Lower Trays, Analogs, Copings, Digital Scan No., Tooth No., and 3-zone **Shade Master (Gingival, Body, Incisal)** + photos.
- **3-Stage Pipeline:** *Ready to Dispatch → Check Work (with overdue red alert) → Work Ready* (with patient recall statuses Schedule / No Ans 1–3).
- **Contextual 3-Lane Chat:** Maximize/minimize chat drawer per lab job featuring dedicated *Doctor, Reception, and Lab* communication lanes.

9. Insurance Tracker & TPA Submission

INS-01 to INS-10

- **3-Stage Claim Lifecycle:** Tracks claims across *Not Submitted, Submitted, and Payment Received* with multi-procedure invoice grouping and insurer `Pay` reconciliation.
- **Insurer-Specific Submission Formats:** Dynamic templates for **Globemed** (includes Signed Invoice column) and **Mednet** (uses Mednet Auth No.) with 6 per-claim document attachments (Invoice PDF, Claim Form, CPR, Approval, Signed Invoice, X-Ray).
- **3-Decimal BHD Formula Engine:** Enforces `Discount = Gross - Net`, `Deductible = Net × Policy %`, and `Net Receivable = Net - Deductible` with batch PDF, Excel, and Print export.

10. Patient Journey 24-Column Quality Matrix

PJ-01 to PJ-06

- **Chronological Seen-Visit Audit:** Automatically lists all visits marked `Seen` ordered from earliest to latest Seen timestamp.
- **Real-Time Compliance Checklist:** Tracks *Appointment, Account Details, Treatment Plan, General Consent, Treatment Consent, Instruction Sheet, PFE, Financial Agreement, Before/After Photos, Initial Screening, Chief Complaint, FLP Note, Instagram, Google Review, and Feedback*.
- **Actionable Gap Resolution:** Renders completed items in green ticks and missing mandatory items in red crosses with one-click deep links to upload/complete the missing artifact.

11. Billing / Accounts Register & Shift Handover Control

BILL-01 to BILL-05

- **Receptionist-Scoped Cashier Ledger:** Displays bills created under the logged-in receptionist (`Billed By`) with Date Filter and global File/Phone/CPR/Name search.
- **Multi-Tender BHD Breakdown:** Automatically sums numeric **Cash, Credimax, Cheque, and EazyPay** columns to 3 decimal places (`0.000 BHD`) while displaying Insurance Receivables separately and treating Payment Method strictly as a descriptive label.
- **Zero Double-Counting Sync & Signed Printout:** Synchronizes directly with the Dashboard Collection tracker (BILL-05 / CORE-09) and generates a formal shift verification printout with Receptionist Name & Signature blocks.

12. Core System Architecture, Shared Doctor/Admin Console & Offline Sync Engine

CORE-01 to CORE-09 • MS-12

- **Offline-First Resilience & Auto-Sync:** Local caching in SQLite / IndexedDB during network outages, seamlessly syncing with live MySQL backend upon reconnection without data loss or duplicate records.
- **Custom .NET Smart Card Driver Service:** Direct hardware integration via .NET microservice to extract CPR card data into EMRS instantly with 0ms manual typing latency.
- **RabbitMQ High-Throughput Queue:** Asynchronous queuing for bulk WhatsApp, SMS notifications, and heavy audit log dispatches, decoupling background processing from reception UI interactions.
- **Shared Doctor/Admin Operational Console:** Provides the critical feeder hooks (procedure status toggles, tooth numbering, lab job triggers, and admin KPI audit overrides) that fuel all 11 reception operational dashboards.

06 Project Execution Roadmap & Milestone Delivery Schedule

Accelerated 6-Month Core Roadmap + 1-Month Deployment Buffer | 28 Working Weeks (7.00 Months)

To optimize investment velocity and achieve rapid clinical go-live for **Al Rabeeh Dental Center**, the project delivery timeline has been condensed into an **accelerated 6-Month Core Engineering Sprint (24 Working Weeks / 120 Working Days)** followed by a dedicated **1-Month Deployment Buffer & Multi-Branch Pilot (4 Working Weeks / 20 Working Days)**. This delivers the complete 11-module clinical suite plus the shared Doctor/Admin trigger console with zero compromise on quality or security.

Milestone Code	Milestone Title & Blueprint Modules Covered	Key Engineering Deliverables	Effort Allocation	Schedule Timeline	Payment Milestone Breakdown (BD 8,400.000 Offer)
MS-01	**Core Architecture, MySQL, RabbitMQ & Reception Dashboard (Module 1)**	CORE-01..09, ReactJS + PHP MVC + MySQL + RabbitMQ Queues, SQLite/IndexedDB Offline Sync, 6 KPI Cards & 24h Audit	2.5 Wks (0.625 Mo)	Weeks 1 – 2.5 (Month 1)	**15% on Kickoff & MS-01 (BD 1,260.000)**
MS-02	**Multi-Doctor Calendar, .NET CPR Driver & Bulk WhatsApp (Module 2)**	CAL-01..12, Drag-Drop Scheduler, 20 Status Counters, Custom .NET Smart Card Driver & Service, WhatsApp/SMS Queue	2.5 Wks (0.625 Mo)	Weeks 2.5 – 5.0 (Month 1.25)	**10% on MS-02 UAT (BD 840.000)**
MS-03	**Missing Schedule (MSD) Leakage Tracker (Module 3)**	MSD-01..05, Automated Unbooked Patient Engine, 7 Clinical & Balance Category Cards, Instant Booking Removal	1.5 Wks (0.375 Mo)	Weeks 5.0 – 6.5 (Month 1.6)	**15% on MS-03 & MS-04 UAT (BD 1,260.000)**
MS-04	**Fail Visit Rollover & No-Show Recovery (Module 4)**	FV-01..09, Yesterday Unattended Rollover, Deduplication, 10 Call Status Filters, Call History Drawer, Booking Lock	2.0 Wks (0.500 Mo)	Weeks 6.5 – 8.5 (Month 2.1)	
MS-05	**Ortho Continuity & Overdue Aging Tracker (Module 5)**	ORTHO-01..08, Doctor Running PT Filter, 5 Appliance Cards, 1–6+ Month Aging Buckets, Ortho Call List	2.0 Wks (0.500 Mo)	Weeks 8.5 – 10.5 (Month 2.6)	**15% on MS-05 & MS-06 UAT (BD 1,260.000)**
MS-06	**Treatment Plan (TP) & Tooth-Level Recall Tracker (Module 6)**	TP-01..10, 3-Rule Active Filter, BHD TP Value Metrics, Favorite TP, FDI Tooth Recall (Scaling, GBT, RCT, Extraction)	2.0 Wks (0.500 Mo)	Weeks 10.5 – 12.5 (Month 3.1)	
MS-07	**Post-Treatment Feedback Call Tracker (Module 7)**	FB-01..06, Procedure Completion Trigger, Live Call Percentage Funnel, Dual Doctor/Reception Comment Lanes	1.5 Wks (0.375 Mo)	Weeks 12.5 – 14.0 (Month 3.5)	**15% on MS-07 & MS-08 UAT (BD 1,260.000)**
MS-08	**Dental Lab Lifecycle, Shade Master & 3-Lane Chat (Module 8)**	LAB-01..10, Job Order & 3-Zone Shade Master, 3-Stage Pipeline, Overdue Alert, 3-Lane Chat + External Lab Link	2.0 Wks (0.500 Mo)	Weeks 14.0 – 16.0 (Month 4.0)	
MS-09	**Insurance Tracker & Globemed/Mednet Submission (Module 9)**	INS-01..10, Multi-Procedure Invoices, Deductible/Receivable Engine, 6 Claim Attachments, Globemed/Mednet PDF/XLS	2.5 Wks (0.625 Mo)	Weeks 16.0 – 18.5 (Month 4.6)	**10% on MS-09 UAT (BD 840.000)**
MS-10	**Patient Journey 24-Point Compliance Matrix (Module 10)**	PJ-01..06, Chronological Seen Audit, 17+ Green/Red Compliance Checkpoints, Tablet E-Consent & Photo Links	1.5 Wks (0.375 Mo)	Weeks 18.5 – 20.0 (Month 5.0)	**10% on MS-10 & MS-11 UAT (BD 840.000)**
MS-11	**Billing / Accounts Register & Shift Handover (Module 11)**	BILL-01..05, Receptionist Cashier Register (Cash, Credimax, Cheque, EazyPay), Collection Auto-Sync, Signed Printout	2.0 Wks (0.500 Mo)	Weeks 20.0 – 22.0 (Month 5.5)	

Milestone Code	Milestone Title & Blueprint Modules Covered	Key Engineering Deliverables	Effort Allocation	Schedule Timeline	Payment Milestone Breakdown (BD 8,400.000 Offer)
MS-12	**Shared Doctor & Admin Trigger Console + Pre-Go-Live Integration**	Doctor Encounter/Procedure/Tooth/Lab Triggers, Admin KPI Approval & Task Console, VAPT Security Audit & Training	2.0 Wks (0.500 Mo)	Weeks 22.0 – 24.0 (Month 6.0)	**5% on Core Completion (BD 420.000)**
SUBTOTAL: 6-MONTH ACCELERATED CORE DEVELOPMENT (MS-01 TO MS-12)			**24.0 Wks (6.00 Mo)**	**Weeks 1 – 24 (Months 1 – 6)**	**BD 7,980.000 (95% Payment Subtotal)**
MS-BUF	**Dedicated Deployment Buffer, Multi-Branch Pilot & Hypercare Stabilization**	**Extra 1-Month Buffer:** Multi-branch pilot across 6 Al Rabeeh clinics, load testing, contingency buffer, end-user hypercare	**4.0 Wks (1.00 Mo)**	**Weeks 25 – 28 (Month 7)**	**5% on Final Sign-off (BD 420.000)**
TOTAL ACCELERATED ENGAGEMENT (6 MONTHS CORE + 1 MONTH BUFFER EXTRA)			**28.0 Wks (7.00 Mo)**	**Months 1 – 7 (28 Working Wks)**	**BD 8,400.000 (100% Total)**

07 Commercial Investment & Resource Cost Analysis

Dedicated 7-Member Delivery Organization (BHD Standard)

Our commercial model is anchored on a dedicated **7-member cross-functional delivery organization** (5 Full-Time Core Engineering Squad members + 2 Executive Steering & Security Advisory specialists) at a dedicated consolidated commercial investment rate of **BD 1,200.000 / Month (BD 300.000 / Week)**. This delivers the complete 6-Month Core Engineering Sprint at **BD 7,200.000** (24 Working Weeks) and the dedicated 1-Month Deployment Buffer & Hypercare at **BD 1,200.000** (4 Working Weeks), for a single consolidated **Total Accelerated Contract Offer of BD 8,400.000** across the 28-week engagement.

7.1 Dedicated Engineering Squad & Advisory Rate Card

Designation / Domain Role	Allocation	Monthly Rate (BHD)	Weekly Rate (BHD)	Hourly Rate (160h/mo)
A. CORE DEDICATED ENGINEERING SQUAD (5 FULL-TIME ENGINEERS)				
1. Project Manager & Scrum Delivery Lead	100% Dedicated	**BD 409.091**	BD 102.273	BD 2.557
2. Lead Backend & Middleware Architect (PHP MVC / Microservices / RabbitMQ)	100% Dedicated	**BD 204.545**	BD 51.136	BD 1.278
3. Lead Frontend & Offline Sync Engineer (ReactJS / SQLite / IndexedDB)	100% Dedicated	**BD 204.545**	BD 51.136	BD 1.278
4. Database, Queue & .NET Hardware Driver Engineer (MySQL / .NET CPR Service)	100% Dedicated	**BD 172.727**	BD 43.182	BD 1.079
5. Senior QA & Automated Testing Engineer	100% Dedicated	**BD 127.273**	BD 31.818	BD 0.795
Subtotal: Core Engineering Squad (5 Members)	**800 Man-Hours / Mo**	BD 1,118.182	**BD 279.545**	**BD 6.989**
B. EXECUTIVE STEERING & SECURITY ADVISORY TIER (2 SPECIALISTS)				
6. Executive Steering & C-Level Governance Director	Sprint Governance	**BD 50.000**	BD 12.500	BD 0.313
7. Enterprise Security, Encryption, VAPT & Cloud Governance Specialist	AES-256 / NHRA / VAPT	**BD 31.818**	BD 7.955	BD 0.199
Subtotal: Executive & Domain Advisory Tier (2 Specialists)	**Continuous Oversight**	BD 81.818	**BD 20.455**	**BD 0.511**
TOTAL CONSOLIDATED PROJECT DELIVERY MATRIX (7 MEMBERS)	**Full Squad + Advisory**	BD 1,200.000 / Mo	BD 300.000 / Wk	BD 7.500 / Hr

7.2 Milestone & Module-Wise Commercial Investment Breakdown (BHD)

Milestone & Module Reference	Effort Duration	Core Squad Cost (BHD)	Advisory Cost (BHD)	Total Module Cost (BHD)
MS-01 (Module 1): Core Shared Engine, Offline Sync & Reception Dashboard + 6 KPI Trackers	2.5 Wks (0.625 Mo)	BD 698.864	BD 51.136	**BD 750.000**
MS-02 (Module 2): Multi-Doctor Calendar, Smart CPR Reader & Bulk WhatsApp	2.5 Wks (0.625 Mo)	BD 698.864	BD 51.136	**BD 750.000**
MS-03 (Module 3): Missing Schedule (MSD) Automated Leakage Tracker	1.5 Wks (0.375 Mo)	BD 419.318	BD 30.682	**BD 450.000**
MS-04 (Module 4): Fail Visit Unattended Rollover & Follow-Up Engine	2.0 Wks (0.500 Mo)	BD 559.091	BD 40.909	**BD 600.000**
MS-05 (Module 5): Orthodontic (Ortho) Continuity & Overdue Aging Tracker	2.0 Wks (0.500 Mo)	BD 559.091	BD 40.909	**BD 600.000**
MS-06 (Module 6): Treatment Plan (TP) Value & Tooth-Level Recall Tracker	2.0 Wks (0.500 Mo)	BD 559.091	BD 40.909	**BD 600.000**
MS-07 (Module 7): Post-Treatment Patient Feedback Call Tracker	1.5 Wks (0.375 Mo)	BD 419.318	BD 30.682	**BD 450.000**
MS-08 (Module 8): Dental Lab Lifecycle, 3-Zone Shade Master & 3-Lane Chat	2.0 Wks (0.500 Mo)	BD 559.091	BD 40.909	**BD 600.000**
MS-09 (Module 9): Insurance Claims Tracker & Globemed/Mednet Submission	2.5 Wks (0.625 Mo)	BD 698.864	BD 51.136	**BD 750.000**
MS-10 (Module 10): Patient Journey 24-Point Real-Time Compliance Checklist	1.5 Wks (0.375 Mo)	BD 419.318	BD 30.682	**BD 450.000**
MS-11 (Module 11): Reception Billing / Accounts Register & Shift Handover	2.0 Wks (0.500 Mo)	BD 559.091	BD 40.909	**BD 600.000**
SUBTOTAL: 11-MODULE BLUEPRINT BASELINE (MS-01 TO MS-11)	**22.0 Wks (5.50 Mo)**	**BD 6,150.000**	**BD 450.000**	**BD 6,600.000**
MS-12 (Essential Bridge): Shared Doctor & Admin Operational Console, Tooth Chart & Integration	2.0 Wks (0.500 Mo)	BD 559.091	BD 40.909	**BD 600.000**
SUBTOTAL: 6-MONTH ACCELERATED CORE DEVELOPMENT (MS-01 TO MS-12)	**24.0 Wks (6.00 Mo)**	**BD 6,709.091**	**BD 490.909**	**BD 7,200.000**
MS-BUF (Deployment Buffer): Extra 1-Month Multi-Branch Pilot & Stabilization Buffer	4.0 Wks (1.000 Mo)	BD 1,118.182	BD 81.818	**BD 1,200.000**
TOTAL RECALCULATED CONTRACT VALUE (6 MO CORE + 1 MO BUFFER)	**28.0 Wks (7.00 Mo)**	**BD 7,827.273**	**BD 572.727**	**BD 8,400.000**

7.3 Production Server Provisioning, Data Migration & Hardware Driver Schedule

Solution Reference	Scope & Technical Specification	Engagement / Billing Model	Commercial Terms (BHD)
SL-ARDC-SRV-001: Production Cloud / Dedicated Server Infrastructure	Live Server hosting environment for PHP MVC Microservices, MySQL Database & RabbitMQ Broker (with SSL, Daily Backups & Firewall per CORE-08)	**Provided by Client** (Extra charges billed at actuals per year)	`At Actuals / Year (By Client / Server Provider)`
SL-ARDC-CPR-001: Custom .NET Smart Card Driver & Windows Service	Custom .NET Hardware Driver & Background Windows Service to read Bahrain Smart CPR Cards directly into ReactJS	Engineered by SaNDS Lab as part of Milestone 2	`Included in MS-02`
SL-ARDC-MIG-001: Legacy Excel & Patient File Data Migration	Harmonization of existing Patient Files into MySQL, CPR normalization (preserving leading zeros), Active Ortho/TP/Insurance import	One-Time Fixed Scope (3 Weeks)	`BD 1,200.000 (Optional)`

08 Rules, Regulations, Project Governance & Commercial Terms

Mandatory Delivery Governance, Change Control, Server Terms & IP Ownership

To guarantee on-time delivery across all 12 milestones, maintain strict adherence to the agreed blueprint and technology stack, and protect both parties, the following binding governance rules and contractual regulations apply to this engagement:

1. Dedicated Client Project Coordinator (SPOC)

Al Rabeeh Dental Center shall nominate and provide a dedicated **Project Coordinator (Single Point of Contact)** who will be accessible at any time during project execution to provide domain clarifications, master data, clinic workflow confirmations, and consolidated stakeholder sign-offs.

2. Module Approval Timeline & Deemed Confirmation

Upon submitting any module or milestone deliverable for Client Approval (UAT), **SaNDS Lab Middle East W.L.L.** will specify an official review timeline window. The Client must provide consolidated feedback within that stipulated timeframe; **if no feedback is received within the specified period, the module/milestone shall be automatically treated as confirmed and approved**.

3. Revision Limit (Max 3) & Change Request (10% Charge)

Modification or refinement of the same item, screen, or workflow within a delivered module is permitted up to a maximum of **three (3) times**. Any modification requested **more than 3 times on the same item**, or any scope addition outside the signed blueprint, shall be classified as a formal **Change Request (CR)**. **Each Change Request will carry an additional charge equal to 10% of that particular Milestone's value.**

4. 100% Client Source Code & Asset Ownership

Upon project completion and settlement of contract dues, **all custom application Source Code, MySQL database schemas, .NET CPR driver services, APIs, and technical documentation developed under this contract shall be the exclusive asset and intellectual property of the Client (Al Rabeeh Dental Center)**.

5. Strict Blueprint & Technology Stack Compliance

All application modules will be built strictly as per the functional rules defined in the *EMRS Software Blueprint* and the mandated Technology Stack: **ReactJS Front-End, PHP MVC Middleware with Microservices, MySQL Back-End Database, RabbitMQ Server Queuing Services, Custom .NET Smart Card Reader Driver & Service, End-to-End Encryption (AES-256 / TLS 1.3), and Offline-Enabled SQLite / IndexedDB storage with automatic synchronization to the Live Server**.

6. Mutual Contract Termination (60-Day Official Notice)

Either party (**Al Rabeeh Dental Center** or **SaNDS Lab Middle East W.L.L.**) reserves the right to terminate the contract prior to completion by issuing an **Official Written Termination Letter at least sixty (60) calendar days in advance**. All milestones completed or in-progress up to the effective termination date shall be settled pro-rata, with corresponding source code handed over to the Client.

7. Server Infrastructure Provided by Client (Annual Actuals)

The production **Server Infrastructure must be provided by the Client**, with any server, cloud hosting, SSL, or backup subscription charges billed as **extra charges strictly at actuals as given by the Server Provider per year**. Milestone invoices are payable within **seven (7) working days** of milestone confirmation.

8. 60-Day Warranty, NDA & Bahrain PDPL / NHRA Compliance

SaNDS Lab provides **sixty (60) calendar days of complimentary post-Go-Live bug-fix warranty and hypercare support**. Both parties agree to strict mutual confidentiality (NDA) and full compliance with Kingdom of Bahrain Personal Data Protection Law (PDPL) and NHRA medical record governance.

For and on behalf of Client:
Al Rabeeh Dental Center

Authorized Signatory Name & Title
Date & Official Company Stamp

For and on behalf of Engineering Partner:
SaNDS Lab Middle East W.L.L.

Authorized Executive Director
ISO 9001:2015 Certified Engineering Division

SaNDS Lab Middle East W.L.L. × Al Rabeeh Dental Center

Engineered in compliance with Al Rabeeh Dental Center EMRS Blueprint (Revised 19 October 2026) • ReactJS • PHP MVC Microservices • Offline-to-Live Auto-Sync